

POLYMERS – STRUCTURE & PROPERTIES

KEY VOCABULARY

Polymer: a very large molecule made of many small repeating units.

Monomer: the small molecule that joins to make a polymer.

Repeat unit: the part of the polymer that repeats over and over.

Intermolecular force: attraction between polymer chains — fairly strong.

COMMON MISCONCEPTIONS

~~Polymers are made of one giant covalent lattice.~~

✓ They are long molecules held to each other by intermolecular forces.

~~Polymers are gases at room temperature.~~

✓ Forces between long chains make them solids.

~~Monomer and polymer are the same thing.~~

✓ Monomers join to build the polymer.

MONOMER → POLYMER

Many small monomers (e.g. ethene) react together.

The C=C double bond opens up.

Monomers join into one long chain.

$n(\text{CH}_2=\text{CH}_2) \rightarrow -(\text{CH}_2-\text{CH}_2)_n$ (poly(ethene))

The repeat unit is shown in brackets with n.

1 DEFINE POLYMER

State what is meant by a **polymer**. [1 mark]

2 MONOMER

Name the monomer used to make **poly(ethene)**. [1 mark]

3 STATE AT RTP

Explain why polymers are solid at room temperature. [2 marks]

4 DRAW A REPEAT UNIT

Draw the **repeat unit** for poly(ethene). [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Many small ____ join together, as the ____ bond opens, to form a polymer.

monomers

polymer

double

single

6 PROPERTIES

Suggest why polymers do not conduct electricity. [1 mark]